Homework 13: General Knowledge Sweep I

Paper track. This one is not tied to a single session — it covers the loose short-answer items that show up in the "general knowledge / junior detective" section of real invitational tests. Use the Knowledge Document to check yourself. No home lab.

Part A: Powders — uses as identification clues (1 pt each)

Tests often give the **common use** of a powder as a clue. Name the powder (and formula) from its use:

1. Making plaster / casts
2. Antacid / "stomach aches" (fizzes in water)
3. Chalk; also an antacid and a building material
4. Thickening agent (turns iodine black)
5. "Helps with the common cold" (decolorizes iodine)
6. Flavor enhancer / umami
7. Making concrete
8. Making glass
9. Jello / gelatin desserts
10. Bread (tan pellets, biological)

Part B: Flame-test colors (1 pt each)

A flame test can help identify a metal or metal salt. Give the flame color:

1. Copper
2. Sodium (or table salt contamination)
3. Potassium
4. Calcium
5. Lithium / strontium
6. Which two of the six Crime Busters metals are hardest to tell apart by flame color alone, so you'd use density or acid reaction instead?

Part C: Natural vs. synthetic polymers (1 pt each)

1. Name three **naturally occurring** polymers.
2. Name three **synthetic** polymers.
3. Is rayon natural, synthetic, or "regenerated"? Explain.
4. Under a microscope, how do natural fibers usually look different from synthetic ones?

Part D: DNA & PCR (2 pts each)

1. How many chromosomes does a human have? How many pairs?
2. What does RNA stand for, and name one way it differs from DNA.
3. What are the three stages of PCR, in order?
4. What enzyme adds nucleotides to the growing DNA strand during PCR?
5. Why is PCR done *before* gel electrophoresis?
6. On a gel, do longer or shorter fragments travel farther? Why?

Part E: Blood spatter — the information list (5 pts)

"What can you learn from a blood spatter pattern?" List **six** things.

Part F: Blood spatter — drops & velocity (1 pt each)

1. As the height a blood drop falls from **increases**, what happens to the diameter of the stain (up to a point)?
2. Give the rough drop-size and cause for **low**, **medium**, and **high** velocity spatter.
3. A stain 5 mm wide and 10 mm long — angle of impact?

Part G: Minutiae (1 pt each)

Name the minutia from the description:

1. A single ridge splits into two.
2. A ridge starts, then stops, with no connection.
3. A very short ridge, longer than a dot, between two others.
4. A ridge that connects two parallel ridges.
5. A single ridge divides and comes back together, leaving a gap.

6. The triangular formation where ridge flows meet.

Part H: Odds and ends (1 pt each)

1. State Locard's Exchange Principle.
2. What two body structures (besides hair) are also made of keratin?
3. What hormone group regulates human hair growth?
4. What is the study of fingerprints called? Of handwriting?
5. Define "agglutination" as used in blood typing.
6. Name three types of chromatography.
7. In paper chromatography, what two forces make the solvent climb the paper?
8. What is the empirical formula of acetic acid ($C_2H_4O_2$)?
9. How many note sheets may each participant bring, and what size?
10. List five things that can be determined from a shoeprint.